Electronic Supplementary Material 3

Article title: Epidemiology, Clinical Features, and Treatment Modalities of Facial Nerve Palsy in COVID-19 Patients: A Systematic Review.

Journal name: Neurological sciences

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Supplementary Table 3 Detailed inputs for the quality assessment of the included studies

Reference ID	*1. Were the patient's demographic characteristics clearly described?*	*2. Was the patient's history clearly described and presented as a timeline?*	*3. Was the current clinical condition of the patient on presentation clearly described?*	*4. Were diagnostic tests or assessment methods and the results clearly described?*	*5. Was the intervention(s) or treatment procedure(s) clearly described?*	*6. Was the post-intervention clinical condition clearly described?*	*7. Were adverse events (harms) or unanticipated events identified and described?*	*8. Does the case report provide takeaway lessons?*	*Overall quality rating (1-3: poor; 4-6: fair; >6: good)* [a]
Correa/2021/Brazil [4]	1	1	1	1	1	1	0	1	7
Lima/2020/Brazil [15]	1	0	0	1	1	1	0	1	5
Ribeiro/2020/Brazil [18]	1	0	0	1	0	0	0	1	3
Mehta/2020/Canada [16]	1	0	1	1	0	0	0	1	4
Wan/2020/China [21]	1	0	1	1	1	1	0	1	6
Derollez/2020/France [7]	1	0	1	1	1	1	0	1	6
Roussel/2020/France [19]	1	0	1	1	1	1	0	1	6
Decio/2020/Italy [6]	0	0	0	1	1	1	0	1	4
Kumar/2021/India [14]	1	1	1	1	1	1	0	1	7
Karimi-Galougahi/2020/Iran [13]	1	0	1	1	1	1	0	1	6
Homma/2020/Japan [12]	1	0	1	1	1	1	0	1	6
Bastola/2021/Nepal [1]	1	1	1	1	1	1	0	1	7
Dahl/2020/Norway [5]	1	0	1	1	1	1	0	1	6

Figueiredo/2020/Portugal [8]	1	0	1	1	1	1	0	1	6
Goh/2020/Singapore [11]	1	1	1	1	1	1	0	1	7
Neo/2021/Singapore [17]	1	0	1	1	0	1	0	1	5
Cabrera Muras/2020/Spain [2]	1	1	1	1	1	1	0	1	7
Casas/2020/Spain [3]	1	0	1	1	1	1	0	1	6
Garcia Ochoa-Fernández/2021/Spain [9]	0	0	0	1	1	1	0	1	4
Gogia/2020/USA [10]	1	1	1	1	1	0	0	1	6
Theophanous/2020/USA [20]	1	1	1	1	1	1	0	1	7
Zain/2021/USA [22]	1	1	1	1	1	1	0	1	7

[a] Results of the quality checking are 1, 14, and 7 poor, fair, and good articles, respectively.

References

1. Bastola A, Sah R, Nepal G, Gajurel BP, Rajbhandari SK, Chalise BS, Shrestha B, Nepal R, Maharjan K, Dhama K, Rodriguez-Morales AJ (2021) Bell's palsy as a possible neurological complication of COVID-19: A case report. Clinical case reports 9:747-750. https://doi.org/10.1002/ccr3.3631

2. Cabrera Muras A, Carmona-Abellan MM, Collia Fernandez A, Uterga Valiente JM, Anton Mendez L, Garcia-Monco JC (2021) Bilateral facial nerve palsy associated with COVID-19 and Epstein-Barr virus co-infection. European journal of neurology 28:358-360. https://doi.org/10.1111/ene.14561

3. Casas E, Barbosa A, Rubio-Garcia E, Cebrian J, Diaz-Perez C, de la Fuente E, Vivancos J, Lopez-Manzanares L (2020) [Isolated peripheral facial paralysis in a patient with COVID-19]. Revista de neurologia 71:40-41. https://doi.org/10.33588/rn.7101.2020229

4. Correa DG, Hygino da Cruz LC, Jr., Lopes FCR, Rangel CC, de Araujo Henriques Tessarollo AL, Coelho KCG, Brandao RZ, Novis RAF, Novis SSC, Silveira R, Carvalho VT, Nascimento OJM (2021) Magnetic resonance imaging features of COVID-19-related cranial nerve lesions. J Neurovirol 27:171-177. https://doi.org/10.1007/s13365-020-00934-0

5. Dahl EH, Mosevoll KA, Cramariuc D, Vedeler CA, Blomberg B (2021) COVID-19 myocarditis and postinfection Bell's palsy. BMJ Case Rep 14https://doi.org/10.1136/bcr-2020-240095

6. Decio A, Mazza A, Quadri V, Ronconi MS, Brusadelli C, Ruggeri M, D'Antiga L (2021) Neurological Manifestations of COVID-19 in Children: A Case of Facial Nerve Palsy. Pediatr Neurol 116:59. https://doi.org/10.1016/j.pediatrneurol.2020.12.006

7. Derollez C, Alberto T, Leroi I, Mackowiak MA, Chen Y (2020) Facial nerve palsy: an atypical clinical manifestation of COVID-19 infection in a family cluster. European journal of neurology 27:2670-2672. https://doi.org/10.1111/ene.14493

8. Figueiredo R, Falcao V, Pinto MJ, Ramalho C (2020) Peripheral facial paralysis as presenting symptom of COVID-19 in a pregnant woman. BMJ Case Rep 13:e237146. https://doi.org/10.1136/bcr-2020-237146

9. Garcia Ochoa-Fernandez E, Villora-Morcillo N, Taboas-Pereira MA (2021) [Peripheral facial palsy in a paediatric patient with no risk factors within the context of infection by SARS-CoV-2]. Revista de neurologia 72:177-178. https://doi.org/10.33588/rn.7205.2020629

10. Gogia B, Gil Guevara A, Rai PK, Fang X (2020) A case of COVID-19 with multiple cranial neuropathies. Int J Neurosci:1-3. https://doi.org/10.1080/00207454.2020.1869001

11. Goh Y, Beh DLL, Makmur A, Somani J, Chan ACY (2020) Pearls & Oy-sters: Facial nerve palsy in COVID-19 infection. Neurology 95:364-367. https://doi.org/10.1212/WNL.0000000000009863

12. Homma Y, Watanabe M, Inoue K, Moritaka T (2020) Coronavirus Disease-19 Pneumonia with Facial Nerve Palsy and Olfactory Disturbance. Internal medicine (Tokyo, Japan) 59:1773-1775. https://doi.org/10.2169/internalmedicine.5014-20

13. Karimi-Galougahi M, Yousefi-Koma A, Raygani N, Bakhshayeshkaram M, Haseli S (2021) (18)FDG-PET/CT Assessment of COVID-19-Induced Bell's Palsy. Academic radiology 28:144-145. https://doi.org/10.1016/j.acra.2020.11.001

14. Kumar V, Narayanan P, Shetty S, Mohammed AP (2021) Lower motor neuron facial palsy in a postnatal mother with COVID-19. BMJ Case Rep 14:e240267. https://doi.org/10.1136/bcr-2020-240267

15. Lima MA, Silva MTT, Soares CN, Coutinho R, Oliveira HS, Afonso L, Espindola O, Leite AC, Araujo A (2020) Peripheral facial nerve palsy associated with COVID-19. J Neurovirol 26:941-944. https://doi.org/10.1007/s13365-020-00912-6

16. Mehta S, Mackinnon D, Gupta S (2020) Severe acute respiratory syndrome coronavirus 2 as an atypical cause of Bell's palsy in a patient experiencing homelessness. Cjem 22:608-610. https://doi.org/10.1017/cem.2020.418

17. Neo WL, Ng JCF, Iyer NG (2021) The great pretender-Bell's palsy secondary to SARS-CoV-2? Clinical case reports 9:1175-1177. https://doi.org/10.1002/ccr3.3716

18. Ribeiro BNF, Marchiori E (2020) Facial palsy as a neurological complication of SARS-CoV-2. Arquivos de neuro-psiquiatria 78:667. https://doi.org/10.1590/0004-282X20200127

19. Roussel A, Germanaud D, Bouchoucha Y, Ouldali N, Vedrenne-Cloquet M, Castelle M, Baruchel A (2021) Cranial polyneuropathy as the first manifestation of a severe COVID-19 in a child. Pediatr Blood Cancer 68:e28707. https://doi.org/10.1002/pbc.28707

20. Theophanous C, Santoro JD, Itani R (2021) Bell's palsy in a pediatric patient with hyper IgM syndrome and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Brain & development 43:357-359. https://doi.org/10.1016/j.braindev.2020.08.017

21. Wan Y, Cao S, Fang Q, Wang M, Huang Y (2020) Coronavirus disease 2019 complicated with Bell's palsy: a case report. Research Squarehttps://doi.org/10.21203/rs.3.rs-23216/v1

22. Zain S, Petropoulou K, Mirchia K, Hussien A, Mirchia K (2021) COVID-19 as a rare cause of facial nerve neuritis in a pediatric patient. Radiol Case Rep 16:1400-1404. https://doi.org/10.1016/j.radcr.2021.03.063